

Methods Intro

A method is a block of code that performs a specific task. Methods help us avoid writing the same code again and again. so we can say that Methods improve code reusability and readability.

Syntax to define a method :

```
accessmodifier modifier returntype name(formal argument){  
    //instructions  
}
```

For example :

```
public static void addition(int a , int b) // formal args--> Paramters  
{  
    System.out.println("addition method " );  
    int sum = a + b;  
    System.out.println(a + " + " + b + " = " + sum);  
}
```

Explanation

public : The method can be accessed from anywhere in the program.

static : The method belongs to the class and can be called without creating an object.

void : The method does not return any value.

addition : This is the name of the method.

(int a, int b):These are called parameters (formal arguments).

They receive the values passed during the method call.

NOTE :

- A method can't create inside another method.
- A class can have any number of methods.

NOTE :

A method will get executed only when it is called, we can call a method with the help of a method call statement.

Method call statement

The statement which is used to call a method is known as a method call statement.

Syntax to create a method call statement :

```
methodName(Actual arguments);
```

We can call a no-argument method without passing an actual argument in the method call statement.

METHOD CALL STATEMENT FLOW :

- Execution of calling method is paused
- Control is transferred to the called method.
- Execution of called method begins.
- Once the execution of the called method is completed the control is transferred back to the calling method.
- Execution of calling method resumes.